MILITARY SPECIFICATION

BOARD ASSEMBLY, GENERATOR: 10548611

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Generator Board Assembly, 10548611, herein referred to as the generator board, a part of the Digital Generator, 10548613. The digital generator has two applications: a subassembly of Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2 and a subassembly of MX-8048/TPM-22, a part of Radar Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacturing and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-c-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of
MIL-G-50322A(MU)	Generator, Digital: 10548611

STANDARDS

Military

MIL-STD-105	Sampling procedures and Tables for inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., Pa, 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

D10548611	Board Assembly
D10548614	Schematic, Master Clock
D10548613	Generator, Digital

PACKAGING DATA SHEET

10548611	Packaging of Board Assembly

(Copies of specifications, standards, drawings and packaging data sheets rewired by the suppliers in connection with specific procurement functions should be obtained-from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The generator board shall be manufactured in accordance with Drawing D10548611 and drawings pertaining thereto.

3.2 General specifications. - Where applicable, the generator board shall meet the requirements of MIL-F-13926 and MIL-c-46357 as follows":

3.2.1 MIL-F-13926.

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishings

d. Part identification and marking

e. Workmanship

f. Soldering electrical connections

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3.2.2 MIL-C-46357.

a. Etched circuit assemblies

b. Wiring

c. Threaded fasteners

d. Markings

e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration,. - The generator board shall be mounted on the vibration table in normal service position. The generator board shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for two hours along each of three mutually-perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The generator board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The half sine shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ± 1 milliseconds (ins). The G value of each impulse shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. - The generator board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The generator board shall meet the requirements of 3.4 and 3.5 after exposure for not less than one hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg.).

3.3.4 Temperature.

3.3.4.1 Operating. - The generator board shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

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3.3.4.2 Nonoperating (storage). - The generator board shall not suffer permanent damage when subjected to not less than four hours at each temperature extreme of -65°F and + 160°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). - The generator board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ± 5 percent for a period of not less than 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the applicable portion of the schematic diagram D10548614 for the generator board.

3.5 Performance. - When supplied with the inputs specified in table 1, the generator board shall yield the output specified in table I.

TABLE I. - INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	CHARACTERISTIC	TERMINAL CONNECTION
Input voltage	+5.0 ± 0.5 Vdc; Less than 200 mA.	Junction of C_1 and R_{11} (+), board ground (return).
Input signal	2.12 ± 0.37 V pk-pk (into 50 ohms) 60 ± .5 MHZ	Junction of R_2 and R_3 (thru a 16 ohm ± 5% resistor).
Output signal	Square wave; amplitude: 1.8v pk-pk minimum, 4.5V pk-pk maximum into a load of 100 ± 5 ohms; frequency, exactly one-eighth (1/8) of the input frequency.	Junction of 22A pin L and 22B pin K, reference to ground

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as-specified herein. Except as-otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that services conform to prescribed requirements.

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4.1.1 General Provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three generator boards manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. 'The Government reserves the right to witness the contractor's inspection.

4.3 Inspection Provisions.

4.3.1 Submission of Product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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4.3.2.3 Functional tests. - The requirements and tests in table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard-ambient temperature (60°F and 90%).

TABLE II. - CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One generator board shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in table III. Three generator boards shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first.- The control samples shall meet the requirements and tests in table IV.

TABLE III. - SPECIAL SAMPLING

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - Visual
302. General specifications	3.2	MIL-F-13926 MIL-C-46357- Visual

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TABLE IV. - SPECIAL SAMPLING, ENVIRONMENTAL

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
303. Vibration	3.3.1	4.6.1.1
304, Shock	3.3.2	4.6.1.2
305 . Altitude	3,3.3	4..6.1.3
306. Temperature	3.3.4	4.6.1.4
307 . Humidity (nonoperating)	3.3*5	4.6.1.5

4.4.1 Failure of sample, - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Govermnent furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.5.2.3 Test equipment. - In conjunction with 4.5.2.1, the test equipment listed in table V shall be used in performance of the applicable test as specified in 4.6.

TABLE V. - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	RF signal generator	Frequency: 60 ± 0.5 MHz, Output: 2.12 V pk-pk (into 50 ohms).
2.	Oscilloscope, dual channel (with X10 probe)	Bandwidth: Dc to 85 MHz. Deflection factor: 10 mV to 5 V/division. Time base: 20 ns to 5 s/division. Input resistance: 1 megohm shunted by 15 pF. Rise time: Better than 10 ns.
3.	Power supply	Output voltage: 0-5 Vdc. Output current: 300 mA. Ripple voltage: Less than 10 mV rms.
4.	Frequency counter	Range: 0-100 MHz. Gate time: 1 µs to 10 s. Accuracy: ± 1 count ± time base accuracy.
5.	Digital voltmeter	Range: 0-10 Vdc. Accuracy: ± 0.1%.
6.	Termination	50 ohms with a VSWR ≤ 1.15:1.
7.	R_1	Resistor, 16 ohm- ± 5%, 1/8 watt.
8,	R_2	Resistor, 100 ohm ± 5%, 1/8 watt.

Note: Wire, cable, adapters, as required.

4.6 Test methods and Procedures.

4.6.1 Environmental. - Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

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4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I of MIL-STD=810 except for conditions specified in 3.3.1 and that failure criteria do not apply. After vibration the generator board shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. - This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except for conditions specified in 3.3.2. After the shock pulses, the generator board shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating. - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in step 2 the altitude shall be limited to 40,000 feet (5.5 inches Hg). Operation of step 3 shall be the performance tests specified in 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). - This test shall be performed-as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of step 2 shall be ± 160°Y. The temperature of step 3 shall be maintained for not less than-four hours. The temperature of step 4 shall be +155°F. In step 5 the temperature shall be maintained for four hours and the generator board shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4,6.1,4.2 Low (operating and nonoperating). - This test shall be performed as specified in Procedure 1, Method 502 of MIL-STD-810. The temperature of step 2 shall be -65°F maintained for not less than four hours. The temperature of step 4 shall be -40°F. In steps 5 and 7, the generator board-shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating. - This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The generator board shall be tested in accordance with 4.6.3.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - Connect the generator board as shown in figure 1.

4.6.3.1 Input voltage. - Adjust the power supply (item 3) for +5.0 ± 0.1 Vdc at the input terminal of the generator board using the DVM (item 5). (Refer to figure 1).

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4.6.3.2 Input signal. - Adjust the rf signal generator (item 1) for: 60.00 MHz with an output level of 2.12 V peak-to-peak into 50 ohms (item 6); using a 16-ohm (item 7) resistor in series, connect the rf signal generator to the input terminal of the generator board. (Refer to figure 1).

4.6.3.3 output signal. - Synchronize the oscilloscope (item 2) to display the input 60 MHz signal on channel 1. With the output loaded with 100 ohms, verify that the frequency displayed on channel 2 is exactly one eighth (eight periods to one) of the input frequency on channel 1. Verify that the output is a square wave with an amplitude of 1.8 to 4.5 V peak-to-peak.

4.6.3.4 Input voltage variation. - Vary the input voltage from +'+.5 ± 0.1 Vdc to +5.5 ± 0.1 Vdc while observing the output. Verify that the output meets the requirements of 4.6.3.3.

4.6.3.5 Input signal frequency variation. - Vary the input frequency from 59.90 to 60.10 MHz while observing the output. Verify that the output meets the requirements of 4.6.3.3. With the input frequency get to 60 MHz, reduce the 60 MHz input level to 1.75 ± 0.05 V peak-to-peak. Verify that the output meets the requirements of 4.6.3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing and marking shall be in accordance witj Packaaging Data Sheet 10548611. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The generator board is used in the Digital Generator, 1O548613. The digital generator is used in the Radar Set, AN/VPS-2 to count down the 59.936 MHz intermediate frequency (i-f) from the STALO (stabilized local oscillator) to approximately 7.5 MHz, The 7.5 MHz output is used in the Computer, CP-888/VPS-2 to initiate timing signals necessary for computation of the range and range rate analog information oucputs.

The digital generator is used in radar test set, subassembly MX=8048/TPM-22 to produce similar VPS-2 radar signal and test signals for testing of a Receiver, R-1475/VPS-2.

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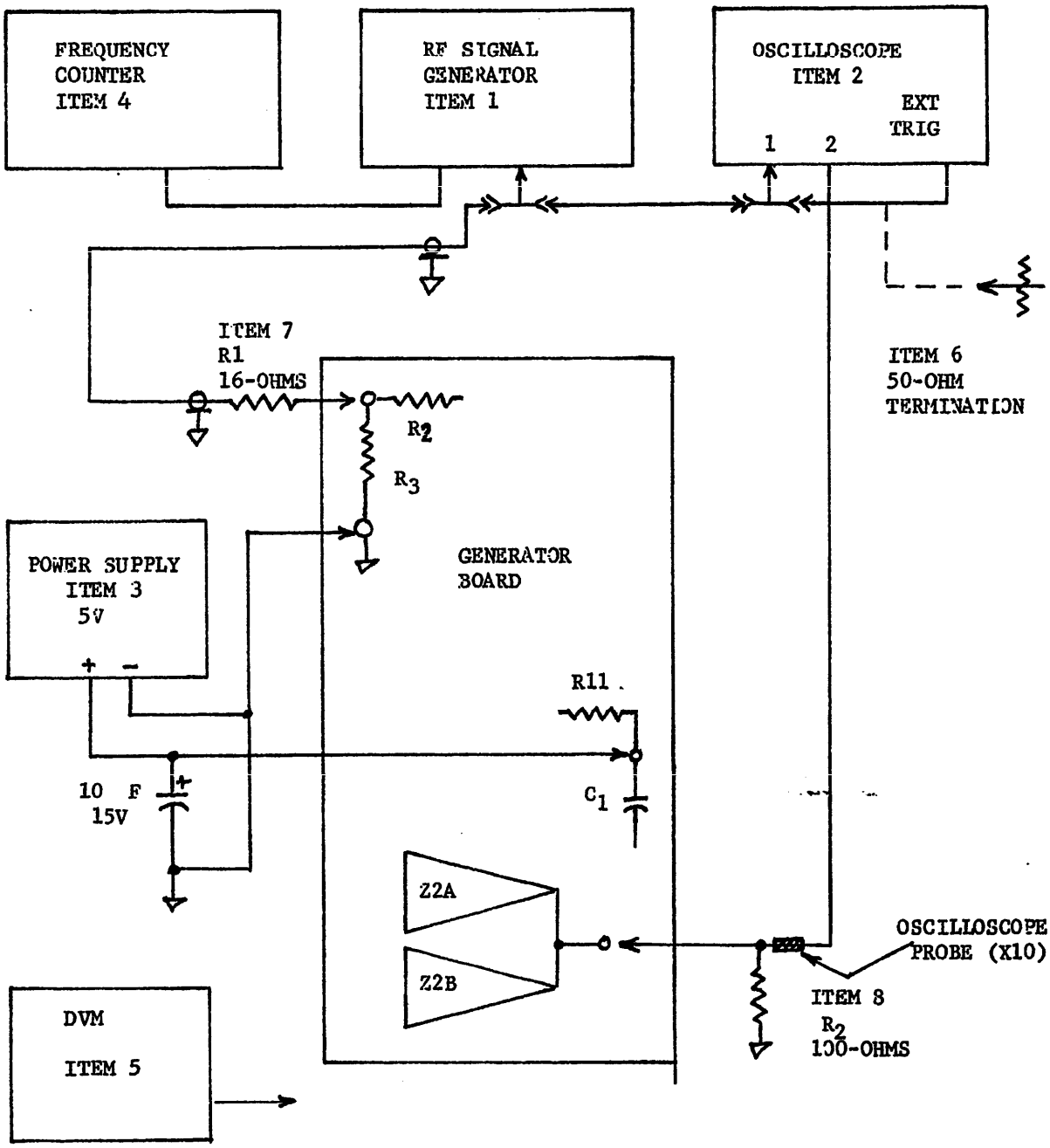


Figure 1. - Test setup

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6.2 Ordering data. - Procurement documents should specify the following :

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project NO. 1285-A374

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DOCUMENT IDENTIFIER AND TITLE

MIL-G-48433(MU) Board Assembly, Generator: 10548611

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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